

Data Cleaning



Unclean Data Example.

Identify the ways in which this data is unclean.

Dogs at the Shelter

Name	Breed	Sex	Weight (lb)	Date of birth	Date Admitted	Favourite Toy
Bert	Miniature Schnauzer	M	12	20-Apr-22	29-Jul-22	Ball
Billie	Labrador Retriever	F	60	15-Jun-21	01-Jan-22	Frisbee
Lorde		F				Teddy
Megan	Dalmation	Female	20	17-Jan-65	09-Sept-66	Old sock
Crunch	Chihuahua	M	100070	29-Jul-22	10-Aug-19	Slipper
Billie	Labrador Retriever	F	60	15-Jun-21	01-Jan-22	Frisbee

Missing Values.

Data cleaning solution: drop this row.

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Billie	Labrador Retriever	F	60	15-Jun-21	01-Jan-22	Frisbee

Inconsistent.

Data cleaning solution: replace this value with consistent one.

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Out of date?

Data cleaning solution: drop data, if not timely (this depends on the on task)

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Inaccurate.

Data cleaning solution: replace value with average Chihuahua weight (imputation).

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Imputation - to calculate the missing value based on other observations.

Duplicated.

Data cleaning solution: drop one of the duplicated rows.

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Data duplicates create multiple kinds of problem.

Irrelevant?

Data cleaning solution: drop column, if irrelevant.

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Data Cleaning – What is Unclean Data?



- Incomplete (has values missing)
- Inconsistent
- Inaccurate
- Old / not timely
- Irrelevant



- Complete
- Consistent
- Accurate
- Timely
- Relevant

Data Cleaning



Also called data cleansing, data scrubbing.

Data cleaning focuses on fixing or removing problematic data.

Data Wrangling



Data wrangling focuses on changing the data format by translating "raw" data into a more usable form.

This can often involve reshaping data, or combining datasets.

In reality, people often use these terms interchangeably.